

# Vishal Kumar Allam

Phoenix, AZ | 816-666-2813 | [itsmevishal36@gmail.com](mailto:itsmevishal36@gmail.com) | **Work Authorization: H1B**

## PROFESSIONAL SUMMARY

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- Senior AI and Software Engineering professional with **5+ years of experience** designing, building, and shipping production-grade AI, backend, and data systems using Python, cloud-native architectures, distributed systems, APIs, and modern software engineering practices.
- Hands-on experience delivering **LLM-powered product features, agentic AI systems, RAG pipelines, autonomous workflows, tool/function calling, contextual reasoning, state management, and human-in-the-loop controls** for real enterprise users.
- Strong backend engineering expertise with **Python, Go, TypeScript, FastAPI, REST, gRPC, microservices, Kafka, event-driven architectures, data pipelines, authentication, validation, and scalable system design**.
- Architected production-grade **retrieval and agent orchestration systems** using embeddings, vector storage, prompt engineering, structured schemas, contextual memory, evaluation frameworks, and monitoring to support accurate and reliable AI behavior.
- Experienced owning production systems across **reliability, latency, cost, observability, failure modes, logging, tracing, alerting, evaluation, incident response, security, explainability, compliance, and auditability**.
- Strong cloud and infrastructure experience with **AWS, GCP, Azure, Kubernetes, Docker, Terraform, CI/CD, Git, distributed services, and secure-by-default deployment patterns** for enterprise-scale AI platforms.
- Demonstrated product mindset and technical ownership through **customer-focused solution design, collaboration with product and design, AI-assisted development, ambiguity resolution, technical judgment, mentoring, and continuous improvement** within regulated enterprise environments.

## EDUCATION

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**University of Missouri Kansas City**

*"Master of Science" in Computer Science*

Kansas City, MO

Jan. 2023 – May 2024

**Jawaharlal Nehru Technological University**

*"Bachelor of Technology" in Computer Science*

Hyderabad, India

Aug. 2015 - May 2019

## TECHNICAL SKILLS

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- **Programming & Backend:** Python, Go, TypeScript, Java, C#, SQL, FastAPI, Flask, REST APIs, gRPC, Microservices, Async Processing, Distributed Systems, System Design, Data Structures.
- **Agentic AI & LLMs:** LLMs, Agentic AI, AI Agents, RAG, Autonomous Workflows, Tool/Function Calling, Context Reasoning, Agent Memory, State Management, Prompt Engineering, LangChain, LangGraph, LlamaIndex.
- **Retrieval & AI Infrastructure:** Embeddings, Vector Databases, Semantic Search, Retrieval Pipelines, Vector Storage, Agent Orchestration, Structured Schemas, Validation, LLM Services, Model Evaluation, Shared AI Infrastructure.
- **Cloud & Distributed Systems:** AWS, GCP, Azure, Kubernetes, Docker, Kafka, Event-Driven Architecture, Streaming Systems, Cloud-Native Services, Scalability, High Availability, Resilient Systems.
- **Observability & Governance:** Logging, Metrics, Distributed Tracing, Alerting, LLM Evaluation, Reliability Engineering, Latency Optimization, Cost Optimization, Failure Analysis, Security, Explainability, Auditability, Compliance.
- **DevOps & Engineering:** Git, GitHub, Azure DevOps, Jenkins, GitLab CI/CD, CI/CD Pipelines, Terraform, Infrastructure as Code, Code Reviews, Release Management, Production Deployment, Monitoring.
- **AI-Assisted Development & Product Delivery:** Claude Code, Cursor, GitHub Copilot, Agentic Development Workflows, AI-Assisted Debugging, Refactoring, Product Collaboration, Customer Outcomes, Technical Judgment, Regulated Financial Systems.

**Software Engineer**  
**PCMI LLC (Full Time)**

Oct 2024 – Present  
Phoenix, AZ, USA

- Architected and shipped production-grade **LLM-powered and agentic AI applications** using Python, FastAPI, Palantir Foundry/AIP, LangGraph, RAG, REST APIs, and cloud services, automating complex insurance and financial workflows used by enterprise business teams.
- Built agentic systems capable of **reasoning over contextual data, invoking tools, managing workflow state, executing real actions, and recovering from failures** through structured orchestration, retry logic, validation, and human-in-the-loop controls.
- Designed production **RAG pipelines over sensitive financial and transactional data** using embeddings, vector databases, semantic retrieval, metadata filtering, structured schemas, validation, access controls, and auditable source attribution.
- Developed reusable **AI infrastructure and backend services** for LLM access, agent orchestration, tool integration, contextual memory, evaluation, monitoring, and API consumption using Python, REST/gRPC patterns, and distributed microservices.
- Owned production reliability across **latency, model/API cost, logging, tracing, alerting, evaluation metrics, failure modes, incident diagnostics, and performance optimization**, improving stability and predictability of AI-powered workflows.
- Deployed scalable AI services using **AWS/Azure, Kubernetes, Docker, Kafka/event-driven patterns, Terraform, Git, CI/CD, secure authentication, and cloud observability**, supporting reliable operation in regulated enterprise environments.
- Partnered with product, design, operations, and engineering stakeholders to **turn ambiguous business problems into shipped customer-focused AI capabilities**, while using Claude Code, Cursor, and GitHub Copilot for implementation, debugging, refactoring, and multi-step agent development.

**Environments:** Python, FastAPI, Palantir Foundry/AIP, LangGraph, LangChain, LLMs, RAG, Vector Databases, Tool Calling, Agent Memory, State Management, REST APIs, gRPC, Kafka, AWS, Azure, Kubernetes, Docker, Terraform, Git, CI/CD, Evaluation, Observability, Security, Auditability.

**Software Engineer**  
**Client: CVS**

Dec 2023 – Sept 2024  
Phoenix, AZ, USA

- Designed and developed production-grade **AI-enabled healthcare and backend services** using Python, FastAPI, AWS, REST APIs, SQL, and microservices, delivering reliable capabilities for regulated customer and operational workflows.
- Built **LLM-powered RAG and agentic workflows** using embeddings, vector retrieval, prompt engineering, tool/function calling, structured outputs, and contextual state to support intelligent enterprise information processing.
- Developed scalable **data and inference pipelines** for ingestion, transformation, retrieval, enrichment, model execution, and serving using Python, cloud services, event-driven processing, and distributed backend patterns.
- Implemented secure backend integrations using **REST APIs, authentication, authorization, schema validation, state management, AWS IAM, JWT, OAuth2/OIDC, and resilient service contracts** across sensitive healthcare applications.
- Established production observability through **CloudWatch, Splunk, structured logging, metrics, tracing, alerting, LLM evaluations, latency analysis, and incident diagnostics**, improving reliability of AI and backend services.
- Deployed cloud-native workloads using **AWS, Docker, Kubernetes concepts, Kafka/event-driven architecture, Git, CI/CD, and automated release pipelines**, supporting frequent production delivery and controlled operational changes.
- Worked closely with product, architecture, data, and engineering teams to **translate customer needs into scalable AI capabilities**, while applying AI-assisted coding, design reviews, debugging, documentation, and continuous improvement practices.

**Environments:** Python, FastAPI, AWS, REST APIs, Microservices, LLMs, RAG, Embeddings, Vector Databases, Tool Calling, OAuth2/OIDC, JWT, IAM, Kafka Concepts, Docker, Kubernetes Concepts, Git, CI/CD, CloudWatch, Splunk, Tracing, LLM Evaluation, Production Support.

**Software Engineer**  
**Infor**

Sept 2021 – Dec 2022  
Hyderabad, India

- Designed and developed enterprise **backend and distributed software systems** using Python, FastAPI, Flask, REST APIs, SQL, and microservices, supporting scalable business applications across complex operational environments.
- Built reusable service layers with **API contracts, authentication, validation, asynchronous processing, workflow state management, error recovery, and distributed-system patterns**, improving reliability and extensibility across enterprise platforms.
- Engineered scalable data-processing pipelines using **Python, PySpark, SQL, APIs, batch processing, and event-driven architectures** for ingestion, transformation, enrichment, and downstream data consumption.
- Implemented event-driven integration patterns using **Kafka concepts, asynchronous messaging, retries, dead-letter handling, idempotency, and service orchestration**, improving resilience across distributed workflows.
- Built operational visibility through **structured logging, service metrics, health monitoring, tracing concepts, performance analysis, exception diagnostics, and root-cause troubleshooting** across production services.
- Applied modern engineering practices using **Git, code reviews, Jenkins, CI/CD, Docker, Kubernetes concepts, API versioning, technical documentation, and secure deployment practices**.
- Contributed to architectural decisions, reusable frameworks, engineering standards, and cross-team collaboration while **solving ambiguous technical problems, optimizing backend performance, mentoring peers, and improving platform reliability**.

**Environments:** Python, FastAPI, Flask, REST APIs, Microservices, SQL, PySpark, PostgreSQL, MySQL, Oracle, Kafka Concepts, Event-Driven Architecture, State Management, Docker, Kubernetes Concepts, Git, Jenkins, CI/CD, Logging, Monitoring, Distributed Systems.

## Associate Engineer

June 2019 – August 2021

*Virtusa*

*Hyderabad, India*

- Developed enterprise **backend services and distributed applications** using Python, Flask, REST APIs, SQL, and microservices architecture, supporting high-volume business workflows and downstream integrations.
- Built reusable Python components for **API processing, business logic, database interaction, state management, asynchronous workflows, scheduling, exception handling, and configuration management**.
- Engineered data-processing services for **ingestion, transformation, validation, enrichment, database persistence, and downstream delivery** using Python, SQL, PostgreSQL, and MySQL.
- Implemented distributed integration patterns using **REST APIs, event-driven concepts, retries, failure handling, service boundaries, authentication, and structured payload validation** across multi-system workflows.
- Improved operational reliability through **structured logging, monitoring, performance diagnostics, alerting concepts, exception tracking, and root-cause analysis** across backend application components.
- Applied disciplined software engineering practices using **Git, CI/CD, Docker, Linux, version control, code reviews, deployment workflows, documentation, and secure API design**.
- Collaborated with architects, engineers, analysts, and business stakeholders to **translate requirements into scalable software solutions, resolve production issues, optimize processing performance, and contribute reusable platform patterns**.

**Environments:** Python, Flask, REST APIs, Microservices, SQL, PostgreSQL, MySQL, Distributed Systems, Event-Driven Architecture, State Management, Git, CI/CD, Docker, Linux, Authentication, Structured Logging, Monitoring, Performance Optimization, Production Support.

## AWARDS & ACHIEVEMENTS

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- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.